

Advancing Legal Reasoning in *Algerian Law*

RAG · KNOWLEDGE GRAPHS · ARGUMENT MINING → *Citation-faithful legal answers*

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01 The Problem

WHY WHY LEGAL QA IS HARD

1.1 Fluent is not lawful

A legal answer cannot be judged by fluency or plausibility. To be valid it must be **grounded in identifiable sources**, cite the **exact governing articles**, stay inside the **Algerian jurisdiction**, and read each provision **as amended** at the relevant date — and where the law is silent, the only correct answer is a **refusal**. A fluent language model guarantees none of this.

SOURCED · ARTICLE-CITED · JURISDICTION-BOUND · AMENDMENT-AWARE · OR *REFUSED*

1.2 What legal questions demand

بين الأساس القانوني لمبدأ خضوع القاضي للنص التشريعي، وحدود سلطته عند غياب النص...

— demands exact articles — *Civil Code, arts. 1–2*

كيف تغيّرت شروط الخلع في قانون الأسرة بعد تعديل 2005؟

— demands the law as amended — *Ord. 05-02 → Family Code, art. 54*

ABSTAIN

هل يحق لصاحب العمل في الجزائر فصل العامل تامة دون إبداء أسباب (at-will)؟

— a foreign-doctrine trap — *Algerian law is silent; the system must abstain*

verbatim from AlgerianLegalBench (244 questions)

1.3 Where today's LLMs fail

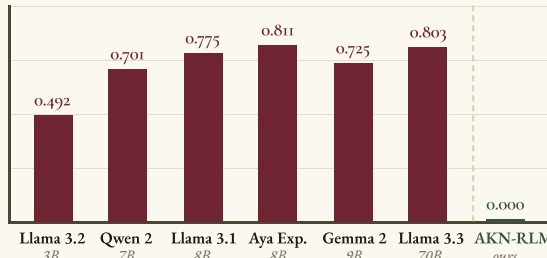


Fig. 1. Hallucinated-citation rate: the share of cited articles that *do not exist*, per direct-LLM baseline (244 questions). Scale does not help — the 70B model fabricates 4 of 5 citations. Gemini 2.5 reaches 0.008 only by declining to cite at all, and six of seven baselines import foreign law on >12 of 40 trap questions. AKN-RLM: 0.000.

1.4 ...and Algerian law raises the bar

- Arabic-first. 232 of 244 benchmark questions are in Arabic
- Layered in time. 1963–2025; amendments rewrite articles in place
- Deeply structured. code → chapter → section → article
- Low-resource. barely covered by existing legal-AI corpora

1.5 Objectives

- ✓ Structure the law
171 docs → Akoma Ntoso XML + legal KG
- ✓ Benchmark it
AlgerianLegalBench: 244 expert questions
- ✓ Reason on evidence
typed retrieval + bounded recursion
- ✓ Verify or abstain
triple faithfulness gate

1.6 Significance & impact

- First citation-faithful QA system for Algerian law — every claim resolves to a registered article, or the system abstains.
- Reusable infrastructure for Algerian legal AI
 - AKN corpus + knowledge graph — 171 structured documents, 1963–2025, article-level registry
 - AlgerianLegalBench — 244 expert-annotated questions, $\kappa = 0.829$
- A recipe for low-resource legal AI — evidence control beats parameter scale.

02 The Solution

HOW STRUCTURED CORPUS → BENCHMARK → AKN-RLM

I Structured legal corpus

the law, made machine-readable

What is Akoma Ntoso? The OASIS LegalDocML standard — an XML vocabulary for legal texts in which every law, article, amendment and cross-reference becomes an **addressable node** under a canonical FRBR URI: the article registry that every citation must resolve against.

WHAT WE BUILT

171 source files
189 MB

45 laws & codes
→ AKN XML

46 RDF graphs
→ one legal KG

1963–2025 coverage,
amendments
in place

```
<!-- Civil Code — Ordinance 75-58, as amended -->
<act name="code_civil">
  <FRBRWork value="/akn/dz/act/1975-09-26/75-58"/>
  <FRBRExpression value=".../ara@2005-06-20"/>
  <lifecycle>
    <eventRef date="2005-06-20" type="amendment"
      source="#law-05-10"/>
  </lifecycle>
  <article eId="art_124">
    <num>124 المادة</num>
    <content>كل فعل أيا كان يرتكبه الشخص بخطئه</content>
    <ref href="#art_127"/>
  </article>
</act>
```



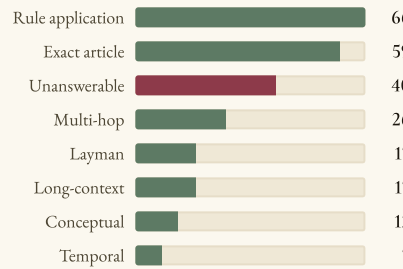
Validated in **multiple manual passes** by two law students and a supervising senior legal expert — the structure is derived from the sources, never invented.

II AlgerianLegalBench

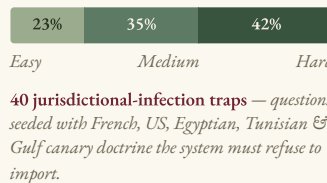
the evaluation layer

244 expert-annotated questions across 23 legal categories — 177 answerable, 67 expecting abstention · 232 Arabic / 12 French · each carries expected articles, a reference answer and a **four-step reasoning chain** · double-annotated, $\kappa = 0.829$.

QUERY TYPES

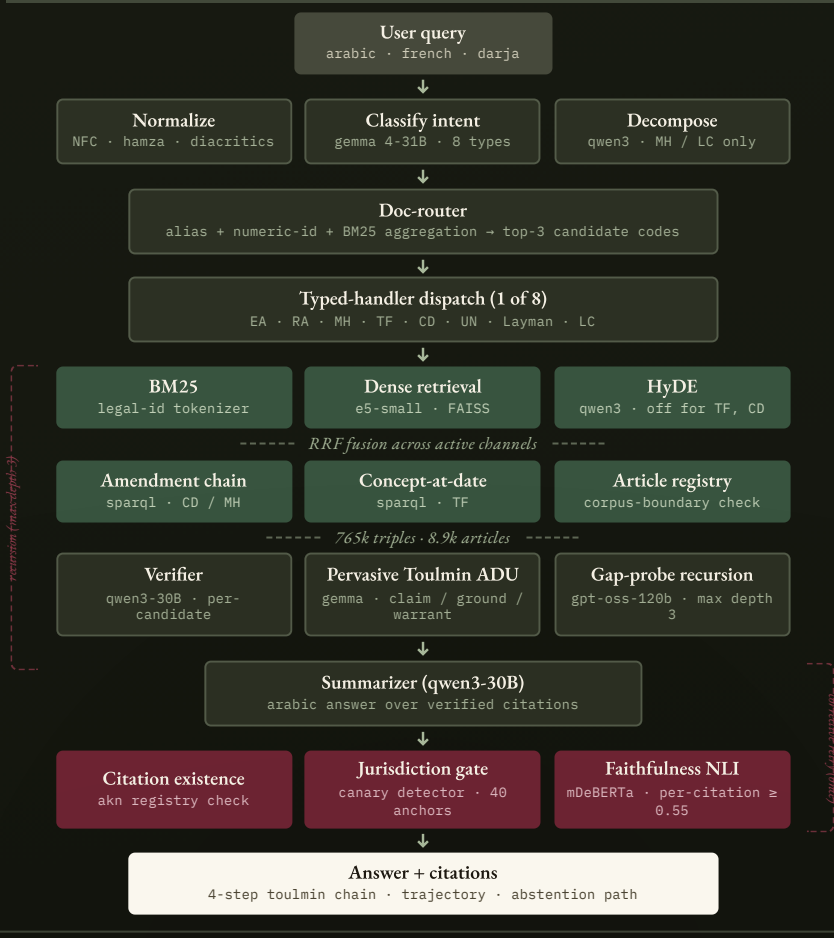


DIFFICULTY



III AKN-RLM

the answer pipeline — Fig. 3.2



03 Results

PROOF FROZEN 244-QUESTION RUN

0.305 Citation F1
locked deployable

✓ 0.000
HCR · JIR

✓ 0.703
Abstention F1

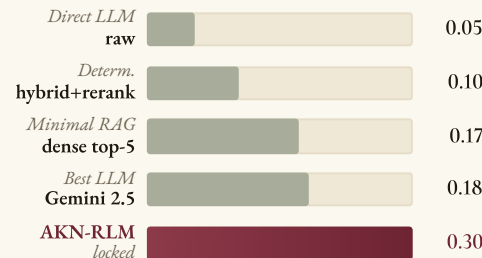
1.64×
vs direct LLM

2.9×
vs determ. retr.

1.74×
vs minimal RAG

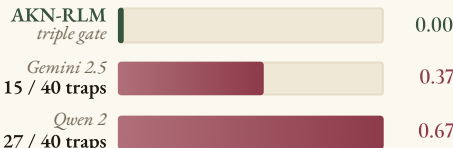
▮ Citation F1 vs baselines

HIGHER = BETTER



✓ Jurisdictional infection (JIR)

LOWER = SAFER



Only AKN-RLM holds **0.000** hallucinated-citation & foreign-law rates. Reliability tracks **legal structure, not scale** — the largest baselines hallucinate most.

△ LIMITS

- Retrieval ceiling: R@10 = 0.258
- French under-served
- Abstention precision 0.654
- Metrics are proxies

🕒 NEXT

- Corpus-tuned embedder
- French ↔ Arabic retrieval
- Graph-retrieval redesign
- Expert legal review



Structured sources



Controlled retrieval



Recursive reasoning



Citation verification

THE TAKEAWAY

Trustworthy Legal AI

an *evidence-controlled system*, not an unconstrained generator.